

Sea Food Allergy

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Introduction

Seafood allergy is an abnormal response of the immune system to proteins in certain aquatic animals. The main risk factors for this allergen are having respiratory problems or having other allergies towards nuts. Typical symptoms include pruritus, itchy throat, swelling, and wheezing when breathing. In severe cases, individuals may experience anaphylaxis, a dangerous reaction that can cause the throat to close. This allergy occurs when the immune system incorrectly identifies proteins in seafood as harmful germs and releases IgE antibodies that trigger the previously mentioned symptoms. The most effective way to prevent allergic reactions is to avoid allergens.

Causes

In an IgE-mediated seafood allergy, the body treats certain seafood proteins as if they were dangerous germs. The immune system produces IgE antibodies, and when the person eats seafood again, those IgE antibodies trigger immune cells to release chemicals that cause symptoms, for example, hives, swelling, or breathing problems (Anvari & Miller, 2019). A major “cause” at the molecular level is which proteins are in the seafood. For shellfish, especially shrimp, crab, lobster, the most important allergen is a muscle protein called tropomyosin. Because tropomyosin is common across many shellfish species, a person allergic to one shellfish may react to others as well. Reviews of shellfish allergy consistently identify tropomyosin as the key allergen and describe other shellfish proteins that can also contribute (Faber et al., 2017; Wong & Huang, 2016).

For fish allergy, the main trigger is often a different protein: parvalbumin. Parvalbumin is a calcium-binding muscle protein that can be a dominant allergen across many fish species, which is why some fish-allergic individuals react broadly (though not always to every fish). Research and reviews repeatedly point to parvalbumin as a major fish allergen and discuss how differences in parvalbumin content and structure can influence reactivity between fish species (Mukherjee et al., 2023).

Another key cause is sensitization-how the allergy develops in the first place. Some people become sensitized after repeated exposure through eating seafood, leading the immune system to start producing IgE against those proteins. Seafood allergy can also be connected to cross-reactivity, where IgE made for a similar protein in another source can “recognize” seafood proteins. A well-known example is the link between house dust mites/cockroaches and shellfish, because they share tropomyosin-like proteins; this can help explain why some individuals with strong dust-mite sensitization show shellfish IgE patterns (Wong & Huang, 2016).

Finally, it’s important to note that not every “seafood reaction” is a true allergy. Some reactions are allergy-like but caused by toxins or chemicals in spoiled fish. For example, scombroid poisoning happens when fish is not kept cold enough and bacteria produce high levels of histamine; meaning the body is reacting to an excessive amount of histamine that has built up in the fish, rather than to an IgE-mediated immune response to fish protein (Centers for Disease Control and Prevention [CDC], 2022; CDC, 2025). This matters because the prevention and follow-up steps can be different even if the symptoms feel similar.

Risk & Factor

Main risk factors for seafood allergies include history of asthma, anaphylaxis, comorbid peanut and nut allergy (the condition of being allergic to both peanuts and tree nuts); allergy to more than 3 seafood allergens can lead to severe seafood allergy. Moreover, having anaphylaxis, comorbid peanut and nut allergy and allergic reactions to more than 3 seafood allergens can signal a potential in shellfish allergy, which is the most widespread kind of seafood allergy. Additionally, gender is considered a minor risk factor. Seafood allergies are more common for adults, and among this population, more are female than male. However, among children, seafood allergies are more common in male.

Symptom & Reaction

Symptoms caused by seafood allergy include a variety of reactions.

- Mild signs can include pruritus(a severe itching of skin), urticaria(itchy bumps on skin), flushing, itchy throat, or rhinoconjunctivitis(inflammation of the rhinitis, appears as swollen eyelids and itchy red eye).
- Moderate signs included sudden swelling of skin, narrowing of airway, coughing, abdominal pain, difficulty breathing (other than wheezing), voice changes, nausea, vomiting, or diarrhea.
- Severe symptoms included wheezing(a high pitched sound produced due to the narrowing of airways in the lung), discoloration of skin/lips/nails and anaphylaxis(severe immune system reaction, causing sudden blood pressure drop and blocking of airway).

Anaphylaxis

Anaphylaxis is a severe allergic reaction that can occur within minutes after exposure to an allergen such as seafood and cause death. Anaphylaxis progresses rapidly, so it is critical to respond without delay, therefore it is important for everyone to recognize the symptoms. Signs such as shortness of breath, wheezing, and coughing are respiratory symptoms. Skin flushing, facial swelling, and itching are also indications that immediate treatments are needed. The first step is to call emergency services such as 9-1-1, because professional medical help is always the safest choice. Epinephrine is the primary treatment option for this allergic reaction. If the person has an epinephrine auto-injector with them, commonly known as an EpiPen, it should be used immediately by injecting it into the outer thigh, neglecting the presence of clothing. Epinephrine works by opening the airways and improving blood circulation, which helps to cancel out the allergic reaction. After getting the injection, that person should be helped into a lying position and kept as still as possible, with their legs slightly elevated to maintain blood flow. It is important to stay with the person to monitor their breathing and be prepared to use a second dose if symptoms do not improve in five minutes and another injector is available.

Prevention

To prevent allergic reactions, it is crucial to stay away from allergens at all times. Avoiding products that contain seafood means to prevent any food intake that has seafood in it, also stay away from touching and inhaling air of cooked seafood, these can also cause allergic reactions. When dining out or shopping for groceries, ask for the ingredients or read labels carefully to avoid seafood; in other cases, try to keep a distance from areas where seafood is prepared and processed.

Additional Information & Hypersensitivity

Type I: immediate response involving IgE antibodies, symptoms include anaphylaxis and asthma.

Type II: IgG/IgM antibodies target antigens on the cell surface

Type III: After seafood proteins enter the body, they may bind with IgG antibodies to form immune complexes. These complexes can circulate in the bloodstream and become deposited in tissue walls, especially in small blood vessels. Once deposited, they activate complement proteins and attract immune cells such as neutrophils. These immune cells release enzymes and inflammatory mediators, which can lead to local inflammation and tissue damage. This type of reaction is slower than typical IgE-mediated food allergy and helps explain how some seafood related immune responses can cause prolonged or systemic symptoms rather than immediate allergic reactions (Abbas et al., 2021, p. 441-443).

Type IV: Type 4 hypersensitivity is the excessive immune response that involves the response of T cells(a type of white blood cell that helps defend the immune system) to stimulus. This reaction involves tissue damage, which is primarily caused by secretion of cytokines(a protein that acts as a messenger to pass out signals to cells), leading to tissue fibrosis (Classification of hypersensitivity reactions 3).

Conclusion

To conclude, seafood allergy is an atypical response of the immune system to specific proteins within marine animals. Individuals with asthma or additional allergies to other foods like tree nuts may be at greater risk. Reactions can range from mild symptoms, like swelling and an itchy throat, to severe and life-threatening. Seafood allergies are caused by the defense cells

that are released when the immune system incorrectly identifies proteins like parvalbumin as a threat. To prevent allergic reactions individuals should avoid seafood and always carry an EpiPen with them just in case an allergic reaction occurs.

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